29.09.2026

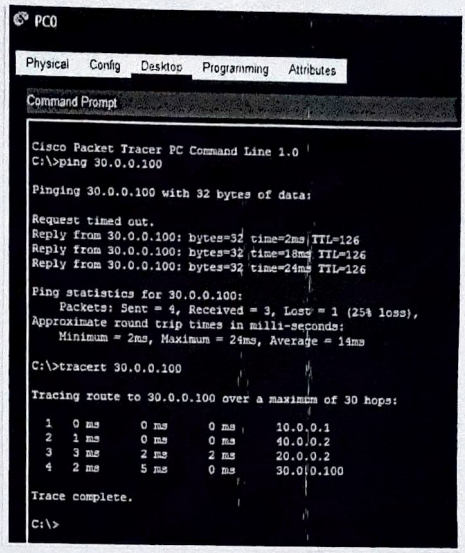
Practical - 11 (a)
Static Routing

AIM:

To simulate and configure static routing with primary backup routes using cisco packet tracer and verify connectivity.

PROCEDURE:

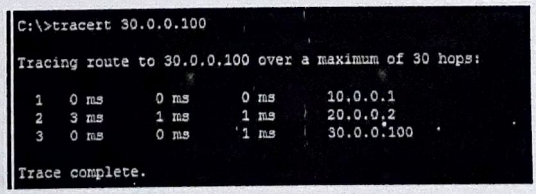
1) Open cisco packet, create topology with Router0, Router1, switch and PCs
2) configure IP address on all routes interfaces and PCs according to given topology.
3) Open CLI of Router0 and enter config mode
4) Static route 30.0.0.0/8 and 50.0.0.0/8 using Router1 as primary route and router2.
5) config separate host route for 50.0.0.100/32 via Enter2 as primary route
6) configure 10.0.0.0/8 and 40.0.0.0/8 with primary and backup routes.
7) The same configuration for Router2
8) use command : show ip route static to verify configured static routes
9) use trace command from a PC to verify path taken by packets to destination.
10) Disconnect primary link between R0 and R1 use tracert again to verify backup route becomes active.
11) Static routes can be removed using : no ip
12) verify routing table again using show ip route static.




```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/1/1
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down
```





```
Router(config-if)#interface serial 0/1/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
```

```
C:\>tracert 30.0.0.100

Tracing route to 30.0.0.100 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms    10.0.0.1
  2    1 ms      7 ms      4 ms    40.0.0.2
  3    9 ms      7 ms      1 ms    20.0.0.2
  4    2 ms      0 ms      4 ms    30.0.0.100

Trace complete.

C:\>
```

RESULT:

Thus the routing with primary and backup routes was successfully configured and simulated using cisco packet tracer.

29.9.2026

Practical - 11(b)

RIP (Routing Information Protocol)

AIM:

To simulate and configure RIP using cisco packet tracer and verify dynamic routing between different networks.

PROCEDURE:

1) Open cisco Packet Tracer and create given topology with R0, R1, R2, PC0, PC1
2) Assign IP address to PCs and router interfaces according to table
3) Configure FastEthernet and serial interfaces of all routers using IP address.
4) Use no shutdown to activate all router interface
5) For DCE serial interface, configure required clock rate and bandwidth.
6) Configure RIP on R0 using:

```
router rip
network 10.0.0.0
network 192.168.1.252
```

7) Configure RIP on R1:

```
router rip
network 192.168.1.244
network 192.168.1.248
```

8) Configure RIP on R2:

```
router rip
network 192.168.1.252
network 192.168.1.244
```

9) Allow RIP to exchange routing information automatically.
10) From PC0, use ping command to verify connectivity with PC1.
11) Use tracert command to observe path followed by packets between networks.
12) Disconnect one of the available routes and use tracert again to verify RIP automatically select the alternate one.

PC1

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126
Reply from 10.0.0.2: bytes=32 time=18ms TTL=126
Reply from 10.0.0.2: bytes=32 time=19ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 19ms, Average = 18ms

C:\>tracert 10.0.0.2

Tracing route to 10.0.0.2 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      20.0.0.1
  2   11 ms     13 ms     12 ms     192.168.1.254
  3   9 ms      5 ms      5 ms      10.0.0.2

Trace complete.

C:\>
```

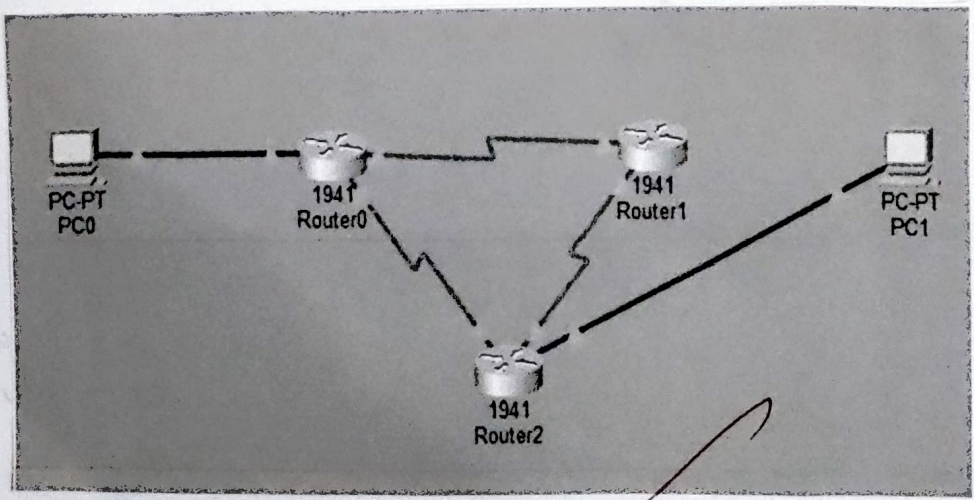
Top

```
C:\> tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    6 ms      8 ms      7 ms      192.168.1.253
  3    5 ms      10 ms     7 ms      20.0.0.2

Trace complete.
```



RESULT:

Thus RIP was successfully configured and simulated using cisco packet tracer.